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Drug Interaction Prediction Using Published Similarity Features

Implementing the logistic-regression component of Cheng and Zhao (2014)

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1. Abstract

This project implements the logistic-regression component of Cheng and Zhao's heterogeneous network-assisted inference (HNAI) framework for drug–drug interaction (DDI) prediction, using the four precomputed drug-pair similarities from their supplementary workbook. A prior-only baseline and Random Forest provide comparisons. The distributed cross-validation worksheet contains 13,952 pairs: 6,946 labeled DDI and 7,006 labeled as sampled non-DDI. In a new stratified five-fold cross-validation, logistic regression achieved a mean area under the receiver operating characteristic curve (ROC-AUC) of 0.654 ± 0.011, within 0.001 of the 0.655 the paper reports for the same classifier. Random Forest achieved 0.676 ± 0.015. A feature ablation reproduced the paper's finding that structural and phenotypic similarity are the most informative inputs. However, both drugs of every test pair also occurred in training. When both drug identities were excluded from training, ROC-AUC was 0.629 (95% bootstrap interval 0.582–0.672) for logistic regression and 0.621 (0.575–0.665) for Random Forest. The scores reflect historical dataset labels and are not clinical risk estimates.

2. Selected paper and research question

The research question is whether similarities between two drugs can distinguish pairs with reported interactions from sampled pairs without reported interactions. Such models may help prioritize combinations for further investigation in therapeutic discovery. The source paper integrates chemical structure, drug targets, therapeutic classes, and adverse-reaction profiles, and evaluates naive Bayes, a decision tree, nearest neighbors, logistic regression, and a support vector machine (SVM) [1]. This project implements logistic regression as the selected existing method and includes Random Forest as an additional comparison.

2.1 Summary of the selected paper

Cheng and Zhao propose heterogeneous network-assisted inference (HNAI), a framework that predicts whether two approved drugs interact from four similarity

measures computed between them [1]. The positive set contains 6,946 clinically reported DDI pairs connecting 721 US Food and Drug Administration (FDA)-approved small-molecule drugs from DrugBank. An equal number of negative pairs was drawn at random from the same 721 drugs, following earlier work; these pairs have no reported interaction but are not confirmed non-interactions.

Each pair is described by four similarities. Phenotypic similarity compares adverse drug reaction (ADR) profiles in a drug–ADR network that combines MetaADEDB with the FDA Adverse Event Reporting System (FAERS). Therapeutic similarity compares Anatomical Therapeutic Chemical (ATC) classifications. Structural similarity compares two-dimensional chemical structures derived from SMILES representations. Genomic similarity, introduced in this paper, is the Tanimoto coefficient between drug–target profiles drawn from DrugBank and the Therapeutic Target Database (2,912 drug–target pairs involving 674 proteins).

The authors trained naive Bayes, decision tree, k-nearest neighbors, logistic regression and support vector machine (SVM) classifiers on these four features. Under five-fold cross-validation, ROC-AUC ranged from 0.565 for the decision tree to 0.666 for the SVM, with logistic regression at 0.655 (Figure 4 of [1]). Performance was poorest when structural or phenotypic similarity was removed. A global SVM model then predicted 81,580 candidate interactions among 729 approved drugs, 559 of which scored above 0.8. In a case study of antipsychotic drugs, 36 predicted interactions scored above 0.8, and several involving weight gain and cytochrome P450 inhibition were supported by published literature.

2.2 Contributions of the selected paper

The paper makes five main contributions [1]:

- A compact framework (HNAI) that represents each drug pair by four similarity scores. The authors acknowledge that some earlier models achieve higher performance, but those rely on high-dimensional features that are harder to interpret.
- A new genomic similarity based on shared drug targets, added to three similarity types from the authors' earlier work.

- A more comprehensive drug–ADR network, integrating MetaADEDB with FAERS reports, for computing phenotypic similarity.
- A systematic comparison of five machine-learning algorithms on identical features, together with a feature-removal analysis identifying structural and phenotypic similarity as the most informative inputs.
- A catalogue of 81,580 predicted interactions with an antipsychotic case study. With the published pair-level features (Supplementary Table S1), this makes the classification stage reusable; the present implementation is built on that release.

2.3 Limitations acknowledged by the authors

The authors identify the lack of confirmed negative interactions, imperfect similarity measurements, and incomplete drug–target and adverse-reaction annotations. They also note that two-dimensional structural similarity can miss three-dimensional relationships and that target profiles omit detailed modes of action and binding information [1]. These limitations constrain the evidence available to the classifiers.

2.4 Critical assessment of the paper

Beyond the limitations the authors acknowledge, this project identifies six weaknesses in the study's design and reporting. Several are supported directly by the experiments in Sections 6–8.

Optimistic evaluation design. Folds were formed from drug pairs, not drugs. In this project's pair-level folds, both drugs of every test pair also appeared in training (Section 6), so cross-validation measures how well a model fills gaps in the interaction network of familiar drugs rather than how it generalizes to new drugs. The paper does not test drug-disjoint splits. When both drug identities were excluded from training (Section 8.1), Random Forest's ROC-AUC fell from 0.676 to 0.621, and the pair-level mean lies above the holdout's 95% bootstrap interval (0.575–0.665).

Unverified and balanced negatives. Negatives were sampled at random from the same 721 drugs in a 1:1 ratio with positives, and some may be unreported interactions. The balance also misrepresents prevalence: the 6,946 known interactions are about 2.7% of the 259,560 possible pairs among the 721 drugs. Because only ROC-AUC is

reported, the precision achievable at realistic prevalence is unknown, and the 0.5 and 0.8 score thresholds used for the global predictions are not calibrated probabilities.

No uncertainty estimates. The paper reports one ROC-AUC per classifier, without variance, confidence intervals or significance tests, yet selects the top-scoring SVM for its global predictions. The gaps between the SVM (0.666), naive Bayes (0.660) and logistic regression (0.655) are at most 0.011, the same size as the fold-to-fold standard deviation observed here (0.011). The ranking of these three classifiers is therefore not established.

Limited contribution of the new genomic feature. The authors describe genomic similarity as an important addition, but in the published data it is zero for 91.5% of pairs and takes only 115 distinct values (Table 2). Removing it produced the smallest loss in the ablation (0.007; Table 6), consistent with the authors' own finding that structural and phenotypic similarity matter most. Sparse target annotation limits what this feature can contribute.

Incomplete reproducibility information. The published materials omit fold assignments, classifier hyperparameters and software versions. The distributed worksheet also disagrees with the manuscript, containing 7,006 rather than 6,946 negatives and 730 rather than 721 drug IDs (Section 3). These gaps prevent exact replication of the reported numbers.

Anecdotal validation of new predictions. The 81,580 predicted interactions were not tested against an independent source, such as a later DrugBank release or clinical records. Support for the antipsychotic predictions comes from selected literature examples, which demonstrates plausibility but not a validated hit rate. Validation with electronic health records is left to future work.

Overall, a ROC-AUC near 0.67 indicates modest discrimination. The framework can help prioritize candidate interactions for further study, but the reported evidence does not support using it to screen drug combinations on its own.

2.5 Connection between the paper and this implementation

- Reused inputs: every row and all four numerical features from the published cross-validation worksheet.
- Implemented component: logistic-regression classification. Training-only standardization, L2 regularization, solver settings, and new stratified folds are specified explicitly in Sections 5 and 6.
- Additional project analyses: a prior baseline, Random Forest comparison, source audit, disjoint-drug holdout, and logistic-regression feature ablation. The source paper also examines feature removal; ablation itself is not claimed as a new method.
- Excluded components: reconstruction of biological features, the other original classifiers including SVM, the global prediction network, and the antipsychotic case study. This is a documented implementation of a selected component, not a complete HNAI reproduction.

3. Published data and observed discrepancy

All rows of the “5-fold CV” worksheet in Supplementary Table S1 were retained [2]. The values in Table 1 were measured directly from the workbook. The manuscript describes 6,946 positive pairs and an equal number of negative pairs involving 721 drugs [1], whereas the distributed worksheet contains 60 additional negative pairs and nine additional drug IDs. The positive subset still contains 721 drugs. The reason for this discrepancy remains unresolved; no rows were removed to force agreement.

Table 1. Counts in the published cross-validation worksheet.

Observed quantity	Count
Published DDI pairs	6,946
Sampled non-DDI pairs	7,006
Total pairs	13,952
Distinct drug IDs in all rows	730
Distinct drug IDs in positive pairs	721

4. Inputs and data integrity

The four numerical inputs are reused directly from the supplementary worksheet. The project implements the classification stage; it does not reconstruct the original drug databases, chemical fingerprints, or biological similarity calculations. Drug IDs and labels are excluded from the feature matrix. Table 2 presents the feature names and the information they represent [1, 2].

Table 2. Published similarity features and their distinct values.

Feature	Information represented	Distinct values
Structural	Chemical structures	1,887
Genomic	Drug target profiles	115
Therapeutic	Anatomical Therapeutic Chemical (ATC) classifications	130
Phenotypic	Adverse-reaction profiles	12,250

SHA-256 hash checks verify the embedded comma-separated values (CSV) file before training. Validation found no missing or nonfinite feature values, duplicate unordered pairs, self-pairs, or constant feature columns. Every value lies within [0, 1]. There are 13,915 distinct feature vectors; one vector is shared by pairs with different labels. These checks establish structural consistency, not medical correctness. In particular, the source label “non-DDI” denotes a sampled negative and does not establish that a combination is clinically safe.

5. Models and preprocessing

Logistic regression applies a linear decision function to standardized similarities, with the logistic link producing a score in [0, 1]. The model uses L2 regularization with C = 1, the lbfgs solver, and a maximum of 2,000 iterations. StandardScaler is fitted only on each training fold within a Pipeline. Random Forest uses 200 trees, a maximum depth of 8, at least 5 training samples per leaf, square-root feature sampling, bootstrap sampling, and a random seed of 42. The prior baseline returns the training class

prevalence. All settings were fixed before evaluation, and no hyperparameter search was performed.

For a pair, the logistic model first forms a weighted sum of the four standardized similarities plus an intercept. Applying the logistic function converts that sum to a score between 0 and 1. Training estimates the weights from source labels. The forest averages class scores across its fitted trees. The threshold predicts DDI when the score is at least 0.5 and predicts the sampled-negative class otherwise; it does not measure interaction severity.

6. Evaluation protocol

The primary experiment uses StratifiedKFold with five folds, shuffling, and a random seed of 42 [3]. All models use identical folds. Each row receives exactly one out-of-fold score from a model trained without that pair. Assertions check complete test coverage, separation of training and test pairs, and scaler statistics calculated from training data only. Both drug identities appear in training for every held-out pair in this run, so the primary experiment evaluates new pairs of familiar drugs.

ROC-AUC measures discrimination across thresholds. Average precision (AP) uses scikit-learn's non-interpolated precision–recall summary [4]. Accuracy, balanced accuracy, precision, recall, and the F1 score use a fixed score threshold of 0.5; precision is reported as zero when its calculation would involve division by zero. The main results tables report fold means and sample standard deviations (SDs). These SDs describe variability across five dependent folds; they are not confidence intervals or significance tests. Pooled curves are reported separately.

The original fold assignments and full software configuration are not available in the worksheet. Differences in data counts, regularization, software, and splitting preclude a claim of exact numerical reproduction. Raw feature construction, the other original classifiers, the global SVM prediction network, and the antipsychotic case study are outside the scope of this project.

7. Results of cross validation for drug pairs

Table 3. Performance across five folds (mean ± sample SD).

Model	ROC-AUC	Average precision	Accuracy	F1
Prior baseline	0.500 ± 0.000	0.498 ± 0.000	0.502 ± 0.000	0.000 ± 0.000
Logistic regression	0.654 ± 0.011	0.641 ± 0.015	0.612 ± 0.011	0.597 ± 0.014
Random Forest	0.676 ± 0.015	0.673 ± 0.016	0.622 ± 0.012	0.603 ± 0.019

Both fitted models rank the dataset labels better than the prior-only baseline. Random Forest has the higher mean ROC-AUC and AP. Its ROC-AUC exceeded that of logistic regression in all five folds, by 0.015 to 0.027 (mean 0.022). Because the folds share training data, this consistency is descriptive evidence rather than a formal significance test.

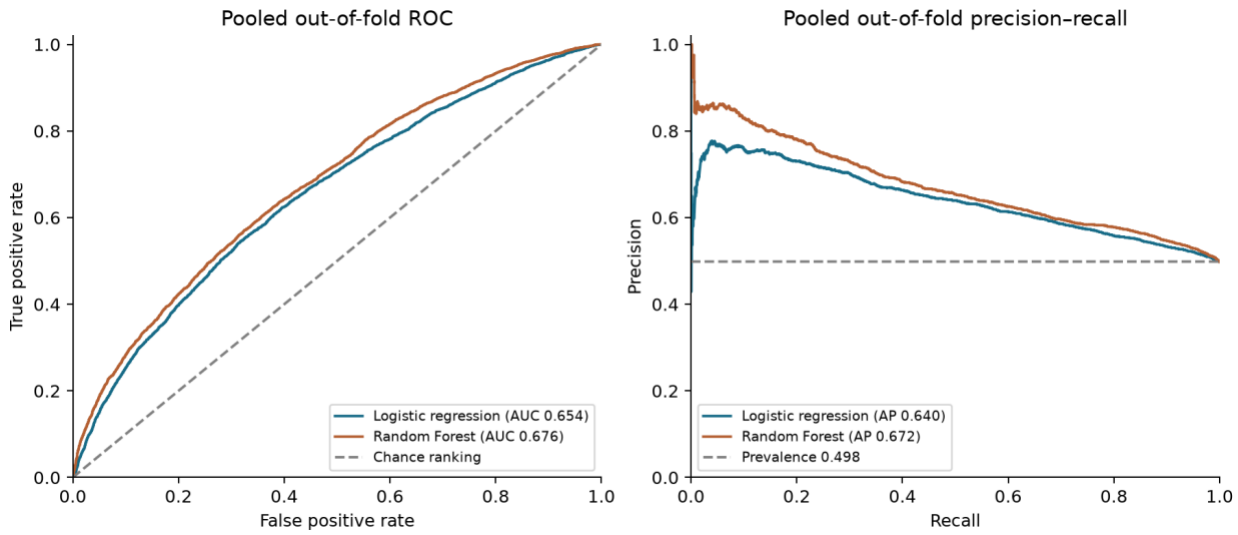


Figure 1. Pooled out-of-fold ROC and precision–recall curves. Each row is scored once while held out. Pooled values differ slightly from the fold averages in Table 3.

7.1 Classification errors at the fixed threshold

At a threshold of 0.5, logistic regression identifies 3,999 of the 6,946 positive labels and misses 2,947. It also predicts DDI for 2,460 of the 7,006 sampled negative labels. Random Forest identifies 4,005 positive labels, misses 2,941, and predicts DDI for

2,330 sampled negative labels. These errors are defined relative to the source labels. Because the negative labels are provisional, the counts do not establish clinical false-alarm rates.

Mean precision and recall are 0.619 and 0.576 for logistic regression and 0.632 and 0.577 for Random Forest. A score threshold of 0.5 therefore leaves substantial misclassification. Accuracy alone would conceal the fact that the prior baseline predicts all pairs as negative and has zero recall for the positive class. Complete fold metrics, predictions, and confusion matrices accompany the notebook.

7.2 Comparison with the original paper

Table 4 compares this implementation with the five-fold results reported by Cheng and Zhao [1]. The logistic-regression ROC-AUC of 0.654 ± 0.011 is within 0.001 of the paper's 0.655, a gap far smaller than the fold standard deviation. This is the central reproduction result: the selected component, rebuilt from the published features with modern software, recovers the published performance. Because the original folds are unavailable and the worksheet contains 60 more negatives than the manuscript describes, the agreement supports a faithful reimplementation but does not prove exact numerical reproduction.

Table 4. Five-fold cross-validation ROC-AUC reported by Cheng and Zhao and obtained in this implementation.

Classifier	Cheng and Zhao (2014)	This implementation
Support vector machine	0.666	—
Naive Bayes	0.660	—
Logistic regression	0.655	0.654 ± 0.011
k-nearest neighbors	0.600	—
Decision tree	0.565	—
Random Forest	—	0.676 ± 0.015
Prior baseline	—	0.500 ± 0.000

Paper values are read from Figure 4 of [1]; no standard deviations are reported. This implementation: mean ± sample SD across five new stratified folds (seed 42). — = not evaluated.

Random Forest's 0.676 is higher than every classifier in the paper. It was not part of the original study, however, and was evaluated on different folds and a slightly different dataset, so this does not establish an improvement over HNAI.

8. Additional evaluation

8.1 Both drug identities excluded from training

A selection using a fixed random seed reserves 146 of the 730 drug IDs. Training uses 8,852 pairs with both endpoints outside the reserved set; testing uses 584 pairs with both endpoints inside it. The remaining 4,516 pairs that cross this boundary are excluded. The test pairs cover 143 reserved IDs and contain 314 DDI and 270 sampled negative labels. An explicit check confirms that no drug identity overlaps between training and testing. All models retain their original settings, and the threshold remains unchanged.

Table 5. Performance on a single exploratory holdout with disjoint drug identities. ROC-AUC intervals are paired 95% percentile bootstrap intervals (2,000 resamples of the 584 test pairs).

Model	ROC-AUC (95% interval)	Average precision	Accuracy	F1
Prior baseline	0.500	0.538	0.462	0.000
Logistic regression	0.629 (0.582–0.672)	0.635	0.587	0.585
Random Forest	0.621 (0.575–0.665)	0.659	0.584	0.577

Both models score lower than in pair-level cross-validation, but a 584-pair test set carries substantial uncertainty. For Random Forest, the pair-level mean (0.676) lies above the holdout's 95% interval (0.575–0.665), suggesting a decline beyond test-set sampling variation. For logistic regression, the pair-level mean (0.654) lies inside its interval (0.582–0.672). The holdout difference between the models is 0.008 in favor of logistic regression, but its paired interval (−0.016 to 0.033) includes zero, so this experiment does not establish which model generalizes better to unseen drugs. Random Forest retains the higher AP. The intervals reflect sampling of test pairs only, not the choice of reserved drugs, and the training size and test population also differ from pair-level cross-validation. The features of held-out drugs are still derived from their published annotations.

8.2 Feature ablation for logistic regression

Each feature was removed separately, and logistic regression was refitted on the same five pair-level folds. The full four-feature model was retained regardless of the ablation results.

Table 6. Logistic regression performance after removing one feature at a time. Changes are computed from unrounded means, so equal rounded means can show different changes.

Feature removed	Mean ROC-AUC ± SD	Change from all four
None	0.654 ± 0.011	—
Structural	0.634 ± 0.008	-0.021
Genomic	0.647 ± 0.012	-0.007
Therapeutic	0.639 ± 0.013	-0.015
Phenotypic	0.634 ± 0.015	-0.020

Removing any single feature reduces mean ROC-AUC in this run. Removing structural or phenotypic similarity produces the largest reductions (0.021 and 0.020), and removing genomic similarity the smallest (0.007). This ordering agrees with Cheng and Zhao, who report that their models performed worst without structural or phenotypic similarity [1], and is a second, qualitative point of reproduction. The results indicate predictive dependence for this model and dataset, not biological causation. The differences between features are small relative to the fold standard deviations (0.008–0.015), and feature correlation further limits their interpretation.

9. Project contributions and limitations

The project contributes a documented modern implementation using the authors' published numerical features, source integrity checks and pair-level validation, a common baseline and model comparison, and additional drug-disjoint and ablation experiments. These are implementation and evaluation contributions for this assignment, rather than a new scientific DDI algorithm. The ablation is an implementation-specific evaluation of an idea also examined in the selected paper.

9.1 Limits of the implementation

Label uncertainty is the central limitation. Unreported interactions may occur among sampled negatives, and the dataset reflects historical coverage rather than current comprehensive knowledge. Missing annotations and unequal study intensity may affect the four similarity measures. Preserving the distributed data makes the experiment traceable but does not resolve the discrepancy with the manuscript counts.

The roughly balanced sample does not represent clinical prevalence. No score calibration, prospective study, or independent clinical validation was performed [5]. Scores cannot be translated into patient-specific risk percentages, and a negative prediction does not establish safety. The symmetric pair features also omit interaction direction, mechanism, severity, dose, timing, and patient characteristics. They support classification, not causal conclusions.

Future work should resolve the discrepancy in the supplementary data, rebuild features from versioned sources, and evaluate repeated drug-disjoint and temporal splits using independently curated interactions. Statistical uncertainty should account for dependence among drug pairs. Broader deployment claims would require evidence beyond the experiments reported here.

10. Conclusion

This project reimplemented the logistic-regression component of Cheng and Zhao's HNAI framework from the authors' published similarity features. It reproduced the paper's reported performance for that component (ROC-AUC 0.654 ± 0.011 versus 0.655) and its finding that structural and phenotypic similarity are the most informative inputs. Under the same pair-level protocol, a Random Forest comparison scored higher (0.676 ± 0.015).

The additional analyses qualify these results. Every test pair in pair-level cross-validation shared both drugs with the training data. Excluding the drugs themselves lowered ROC-AUC to 0.629 for logistic regression and 0.621 for Random Forest, with wide intervals on a single 584-pair test set. Together with unverified negative labels and uncalibrated scores, this indicates that four-feature similarity models can help prioritize

candidate interactions for further study but have not been shown to generalize reliably to new drugs or to support clinical decisions. Repeated drug-disjoint and temporal evaluations against independently curated interactions are the most important next step.

11. Reproduction instructions and submitted artifacts

DDI_Project_Final.ipynb contains the embedded data, explanations, code, and saved outputs. On 15 October 2026, all 11 code cells were rerun in order in a fresh local Python kernel. The matching Python script was also executed separately, and the exported prediction and metric tables were reconciled. The submitted execution used Python 3.14.6, NumPy 2.4.6, pandas 3.0.3, scikit-learn 1.9.0, and Matplotlib 3.11.0. It generated three figures. These saved results are from local execution; this verification did not use a Google-hosted runtime.

The package includes the executed notebook, documented Python source, this report, original supplementary workbook, extracted CSV file, dependency versions, verification scripts, and generated results. The seed, fold assignments, fixed threshold, complete model parameters, file hashes, and runtime versions are recorded. Validation details and a requirement-to-file checklist are in VALIDATION.md and README.md. The commands below use the extracted package directory as the working directory.

11.1 Running the notebook

Upload DDI_Project_Final.ipynb to Google Colab, select a CPU runtime, and run all cells in order. The data are embedded; no Drive connection, API key, GPU, or separate data upload is needed. Wait for the final completion message and inspect the result tables. Download the executed notebook and ddi_results.zip from the runtime. This workflow exports files; it does not submit them to the course.

11.2 Running the source code locally

Use Python 3.14 with the tested dependency file to match the submitted environment. The dependency list pins the runtime packages; a different environment may produce slightly different numerical results. Run these commands from the package directory:

```
python -m venv .venv
```

```
source .venv/bin/activate
python -m pip install -r requirements-tested.txt
python verify_source.py
python ddi_project.py --output-dir rerun_results
python verify_results.py --results-dir rerun_results
```

On Windows, activate the environment with .venv\Scripts\activate instead of the second command. A new output directory preserves the supplied results. The script writes the requested directory and a ZIP beside it. Reusing an output directory replaces generated files with matching names. Use a writable location and run Python normally, without optimization flags that disable assertions.

11.3 Understanding and verifying the outputs

pair_cv_predictions.csv contains one held-out score per model and source pair, with its fold assignment. pair_cv_fold_metrics.csv and pair_cv_summary.csv contain the main evaluation results; drug_holdout_metrics.csv, drug_holdout_auc_bootstrap.csv and ablation_summary.csv contain the additional analyses. provenance.json records the data hashes, settings, versions, and execution times. verify_source.py checks the workbook against the extracted CSV, while verify_results.py independently recalculates exported metrics and checks split membership. SHA256SUMS.json identifies the files supplied in this package.

After running the notebook, inspect_pair("DB00363", "DB00472") retrieves that pair's existing out-of-fold scores. Each score was produced without training on that row, although the drugs themselves occur in other training pairs. The function does not predict an arbitrary unseen pair or identify drug names. The source label and model score are separate quantities: the former records the dataset annotation, while the latter expresses the classifier's support for the DDI class.

12. References

[1] Cheng F, Zhao Z. Machine learning-based prediction of drug–drug interactions by integrating drug phenotypic, therapeutic, chemical, and genomic properties. J Am Med Inform Assoc. 2014;21(e2):e278–e286. https://doi.org/10.1136/amiajnl-2013-002512

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